

Foundations of Residual connection neural network

[Residual connection neural network]

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

1.0 Content Block

Basic block A basic block is the simplest building block studied in the original ResNet. This block consists of two sequential 3x3 convolutional layers and a residual connection. The input and output dimensions of both layers are equal.

2.0 Content Block

Subsequent work Wide Residual Network (2016) found that using more channels and fewer layers than the original ResNet improves performance and GPU-computational efficiency, and that a block with two 3x3 convolutions is superior to other configurations of convolution blocks. DenseNet (2016) connects the output of each layer to the input to each subsequent layer:

3.0 Content Block

Stochastic depth is a regularization method that randomly drops a subset of layers and lets the signal propagate through the identity skip connections. Also known as DropPath, this regularizes training for deep models, such as vision transformers. ResNeXt (2017) combines the Inception module with ResNet. Squeeze-and-Excitation Networks (2018) added squeeze-and-excitation (SE) modules to ResNet. An SE module is applied after a convolution, and takes a tensor of shape $R \times H \times W \times C$ (height, width, channels) as input. Each channel is averaged, resulting in a vector of shape $R \times C$. This is then passed through a multilayer perceptron (with an architecture such as linear-ReLU-linear-sigmoid) before it is multiplied with the original tensor. It won the ILSVRC in 2017.

4.0 Content Block

Residual connection In a multilayer neural network model, consider a (non-residual) subnetwork with a certain number of stacked layers (e.g., 2 or 3). Let $H(x; \alpha)$ denote the subnetwork. Suppose H^* is the desired optimal output of this subnetwork. Residual learning simply adds x directly to the output, such that the optimal learned output now becomes $H^* - x$, which is interpreted as a "residual" with respect to x . The operation of "adding x " is implemented via a "skip connection" that performs an identity mapping to connect the input of the subnetwork with its output. This connection is referred to as a "residual connection" in later work. Let $F(x; \alpha) = H(x; \alpha) + x$. The function F is often represented by matrix multiplication interlaced with activation functions and normalization operations (e.g., batch normalization or layer normalization). As a whole, one of these subnetworks is referred to as a "residual block". A deep residual network is constructed by simply stacking these blocks. Long short-term memory (LSTM) has a memory mechanism that serves as a residual connection. In an LSTM without a forget gate, an input x_t is processed by a function F and added to a memory cell c_t , resulting in $c_{t+1} = c_t + F(x_t)$. An LSTM with a forget gate essentially functions as a highway network. To stabilize the variance of the layers' inputs, it is recommended to replace the residual connections $x + f(x)$ with $x/L + f(x)$,

where L is the total number of residual layers.

5.0 Content Block

where ϕ can be any activation (e.g. ReLU) or normalization (e.g. LayerNorm) operation. This design reduces the number of non-identity mappings between residual blocks, and allows an identity mapping directly from the input to the output. This design was used to train models with 200 to over 1000 layers, and was found to consistently outperform variants where the residual path is not an identity function. The pre-activation ResNet with 200 layers took 3 weeks to train for ImageNet on 8 GPUs in 2016. Since GPT-2, transformer blocks have been mostly implemented as pre-activation blocks. This is often referred to as "pre-normalization" in the literature of transformer models.

6.0 Content Block

Previous work Residual connections were noticed in neuroanatomy, such as Lorente de No (1938). McCulloch and Pitts (1943) proposed artificial neural networks and considered those with residual connections. In 1961, Frank Rosenblatt described a three-layer multilayer perceptron (MLP) model with skip connections. The model was referred to as a "cross-coupled system", and the skip connections were forms of cross-coupled connections.

7.0 Content Block

where L is the index of a residual block and ℓ is the index of some earlier block. This formulation suggests that there is always a signal that is directly sent from a shallower block ℓ to a deeper block L .

8.0 Content Block

This formulation suggests that the gradient computation of a shallower layer, $\frac{\partial E}{\partial x_{\ell}}$, always has a later term $\frac{\partial E}{\partial x_L}$ that is directly added. Even if the gradients of the $F(x_i)$ terms are small, the total gradient $\frac{\partial E}{\partial x_{\ell}}$ resists vanishing due to the added term $\frac{\partial E}{\partial x_L}$.

9.0 Content Block

Degradation problem Sepp Hochreiter discovered the vanishing gradient problem in 1991 and argued that it explained why the then-prevalent forms of recurrent neural networks did not work for long sequences. He and Schmidhuber later designed the LSTM architecture to solve this problem, which has a "cell state" c_t that can function as a generalized residual connection. The highway network (2015) applied the idea of an LSTM unfolded in time to feedforward neural networks, resulting in the highway network. ResNet is equivalent to an open-gated highway network.

10.0 Content Block

Applications Originally, ResNet was designed for computer vision. All transformer architectures include residual connections. Indeed, very deep transformers cannot be trained without them. The original ResNet paper made no claim on being inspired by biological systems. However, later research has related ResNet to biologically-plausible algorithms. A study published in Science in 2023 disclosed the complete connectome of an insect brain (specifically that of a fruit fly larva). This study discovered "multilayer shortcuts" that resemble the skip connections in artificial neural networks, including ResNets.